



t : $\mathbf{v}_A \in VO_{A|B}$ and $\mathbf{v}_B \in VO_{B|A}$; both swerve by the full amount that clears the cone



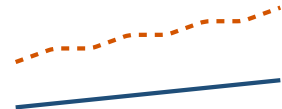
$t + \Delta t$: the other agent's new velocity has moved the cone away; $\mathbf{v}^{\text{pref}}$ is allowed again and both return to it



$t + 2\Delta t$: head-on again; both swerve again, and so on



reciprocal (RVO, ORCA): each agent takes half of the swerve; the relative velocity still clears the cone, and at $t + \Delta t$ nothing has changed, so both keep their velocities



VO: a zig-zag, the velocity changes at every step

ORCA: one smooth swerve